

Development of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a novel Fe-based cathode with high energy densities and excellent cyclability in rechargeable potassium batteries



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ABSTRACT

We introduce $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a novel cathode material with superior electrochemical performance for K-ion batteries. First-principles calculation is used to predict the theoretical properties and detailed K^+ storage mechanism of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, which are consistent with experimental results. $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ exhibits a large specific discharge capacity of $\sim 118 \text{ mAh g}^{-1}$, approaching the theoretical capacity, at C/20 ($1\text{C} = 120 \text{ mA g}^{-1}$) in the voltage range of 2.1–4.1 V (vs. K^+/K), allowing $\sim 3 \text{ mol}$ of K^+ de/intercalation per formula unit with a small volume change of $\sim 4\%$ during charge/discharge. Even at 5C, up to $\sim 70\%$ of its theoretical specific capacity is retained, and this outstanding power-capability is related to the low activation barrier energy for K^+ diffusion, as verified through first-principles calculations. Furthermore, $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ exhibits excellent cyclability with retention of $\sim 82\%$ of the initial capacity after 500 cycles at 5C. The above theoretical and experimental results suggest the feasibility of using $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a cathode material for rechargeable potassium batteries.

1. Introduction

Environmental pollution arising from the usage of fossil fuels has required the development of eco-friendly and sustainable energy storage systems (ESSs) such as rechargeable batteries [1–5]. Currently, lithium-ion batteries (LIBs) are considered one of the best ESSs for large-scale application as well as portable electronics, because of their high energy density and stable cycle life [6–8]. However, despite the many merits of LIBs, current production may not satisfy the explosively growing global demand for LIBs because of the limited reserves and localized distribution of Li resources in the earth's crust [9,10]. Thus, numerous investigations have focused on the development of low-cost ESSs as alternatives to LIBs for grid-scale applications [11–13].

Recently, potassium-ion batteries (KIBs) have attracted significant attention, because of the abundance of potassium resources as well as the lower redox potential of potassium metal (K^+/K : 2.93 V vs. standard hydrogen electrode (SHE)) relative to that of other abundant alkali and rare-earth elements (Na^+/Na : 2.71 V vs. SHE and Mg^{2+}/Mg : 2.27 V vs.

SHE) [14,15]. Furthermore, it has been confirmed that similar to lithium ions, potassium ions can be intercalated into graphite which is currently adopted as a commercialized anode material for LIBs, implying feasibility of KIBs [16–18]. It has known that the cathode materials for KIBs experience slow K^+ ionic diffusion and large structural change during charge/discharge because of the larger ionic size of K^+ ions ($\sim 1.38 \text{ \AA}$) relative to that of Li^+ ions ($\sim 0.76 \text{ \AA}$), which may result in insufficient power-capability and cyclability of KIBs [19,20].

Thus, it is hypothesized that the use of open-structured materials that provide structural stability is a pre-requisite to achieve facile K^+ diffusion and thus long-term cyclability and excellent high-rate performance for KIBs [21]. Herein, we propose a new iron-based mixed-polyanion compound as a suitable cathode for KIBs. First-principles calculation predicts that mixed-polyanion compounds are able to support reversible structural change with moderate ionic diffusion during repeated charge/discharge [22,23]. A high redox potential is an additional benefit originating from the inductive effect of polyanions with high electronegativity [24,25]. Accordingly, we successfully prepared

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$\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, as a novel KIB cathode for the first time, via an electrochemical ion-exchange method. The prediction that the $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox reaction results in a high average operation voltage, ~ 3.21 V (vs. K^+/K), higher than that of any other Fe-based layered oxides reported as a cathode for KIBs [20]. At C/20 ($1\text{C} = 120\text{ mA g}^{-1}$), the electrode delivered a specific discharge capacity of 118 mAh g^{-1} , which is the theoretical capacity corresponding to de/intercalation of 3 mol K^+ per formula unit. At 5C, 70% of the theoretical capacity was delivered, over 82% of which was retained for over 500 cycles, despite the large size of K^+ de/intercalation. Moreover, the use of the earth-abundant Fe element enables the realization of cost-effective production of long-term cyclable high voltage KIBs.

2. Results and discussion

$\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was successfully prepared from $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ using an electrochemical ion-exchange process, as illustrated in Fig. 1a. In case of the direct synthesis of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ using the typical solid-state synthesis methods, not $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ phase but other phases such as KFePO_4 , $\text{K}_2\text{FeP}_2\text{O}_7$ and $\text{Fe}(\text{PO}_3)_3$ were synthesized (Supporting Fig. S1), which indicates the difficulty of preparation of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ through direct synthesis methods. Thus, we performed the electrochemical ion-exchange process to prepare the $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ phase successfully. After deintercalation of 3 mol Na ions from $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (corresponding to the specific capacity of $\sim 120\text{ mAh g}^{-1}$) under the Na-cell system, we performed the electrochemical Na^+/K^+ ion-exchange under K-metal|0.5 M KPF_6 in a EC:PC:FEC = 20:20:1 (v/v)| $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ cell in the voltage range of 2.1–4.1 V for 50 cycles. After the repeated charge/discharge cycling, the voltage profiles are gradually changed (Supporting Figs. S2a and S2b), which indicates the ion-exchange from $\text{Na}_x\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ to $\text{K}_x\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$. The powder X-ray diffraction (XRD) pattern of $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was altered by the electrochemical ion-exchange process in K cells. Comparison of the morphology and elemental ratios of $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ and $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ indicates that all the Na ions in $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ were exchanged to K ions; namely, the elemental ratio of K:Fe in $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was estimated to be $\sim 3.99:3$ by transmission electron microscopy–energy-dispersive X-ray spectroscopy (TEM–EDS) analysis (Fig. 1b and c), which is consistent with the inductively coupled plasma (ICP) spectroscopy results (Supporting Table S1). In addition, scanning electron microscopy (SEM) analyses show that average particle size of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was $\sim 1\text{ }\mu\text{m}$ (Supporting Fig. S3). The successful preparation of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ using the electrochemical ion-exchange process was confirmed through various analyses, such as the XRD analyses with Rietveld refinement, the TEM analyses with EDS-mapping, and the ICP analyses.

Rietveld refinement of the XRD data for the electrochemically ion-exchanged $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ revealed that the $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ exhibited the phase-pure orthorhombic structure with $\text{Pn}2_1\text{a}$ space group, and lattice parameters of $a = 18.4841(6)\text{ }\text{\AA}$, $b = 6.5044(3)\text{ }\text{\AA}$, $c = 11.2670(5)\text{ }\text{\AA}$ (Fig. 2a). Despite $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ and $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ having the same crystal structure, there were differences in the calculated lattice parameters because of ion exchange by the large K^+ ($1.37\text{ }\text{\AA}$) for Na^+ ($1.02\text{ }\text{\AA}$) in $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (Supporting Table S2). Related detailed structural information, including the atomic position, thermal factor, occupancies) is provided in Supporting Table S3. The obtained reliable factors ($R_p = 5.76\%$, $R_l = 6.66\%$, $R_f = 5.04\%$ and $\chi^2 = 2.62\%$) indicate the high accuracy of the Rietveld refinement results for $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$. Considering each atomic position and valence state in the crystal structure, we predicted the possible K^+ sites and diffusion pathways in the orthorhombic $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ structure through bond-valence sum (BVS) energy map analyses using the Bond_Str program in the FullProf package. Fig. 2b and c shows that there are sufficiently large vacant spaces and three-dimensional pathways for K^+ diffusion in the structure, which implies the outstanding power-capability of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a cathode for KIBs. The crystal

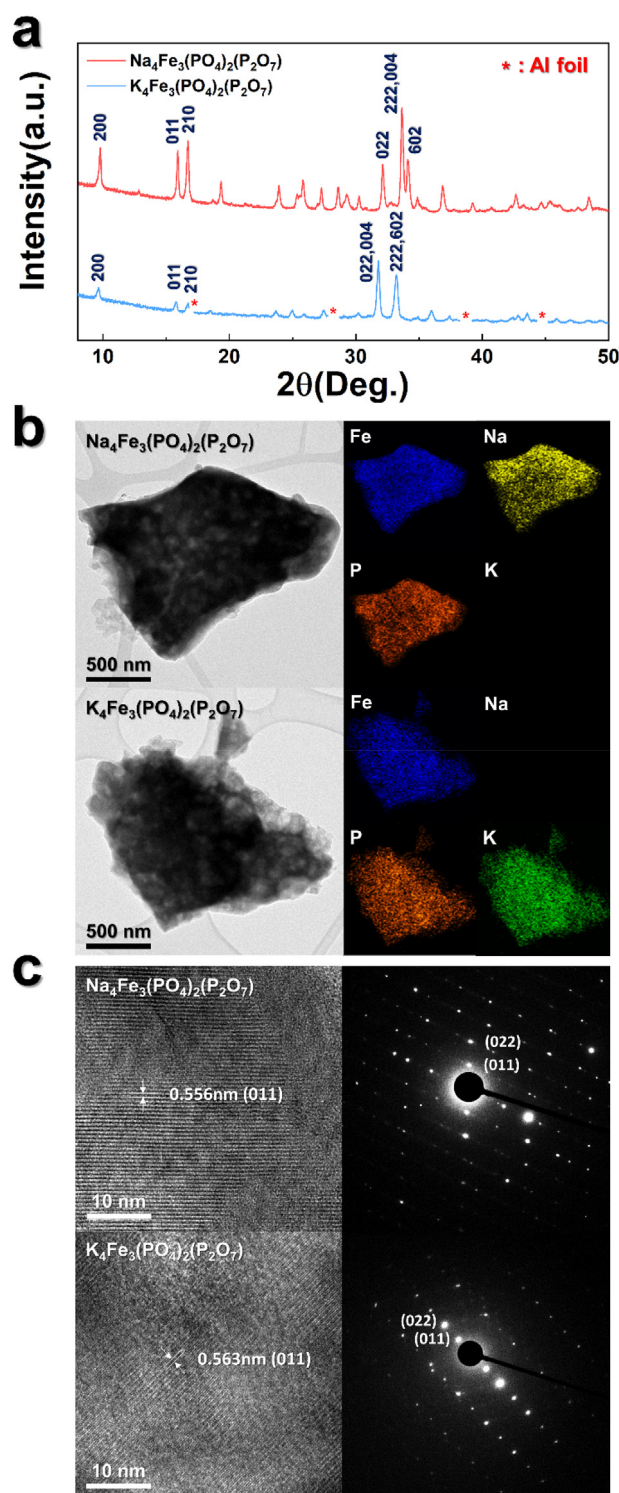


Fig. 1. Characterization of $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ and $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$: (a) XRD patterns, (b) EDS elemental mappings, and (c) magnified TEM images with SAED patterns.

structure of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ consists of the infinite layers of $[\text{Fe}_3\text{P}_2\text{O}_{13}]_\infty$ along the bc plane. In case of $[\text{Fe}_3\text{P}_2\text{O}_{13}]_\infty$ layer, FeO_6 octahedra are connected with P_2O_7 diphosphate along a -axis, which results in the various three-dimensional large pathways for K^+ ionic diffusion in the structure. This crystal structure of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ is similar with those of the other mixed-phosphate $\text{Na}_4\text{M}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ ($\text{M} = \text{Fe, Mn, Co, Ni}$) [25,26]. Each (x, y, z) atomic position on four K sites in

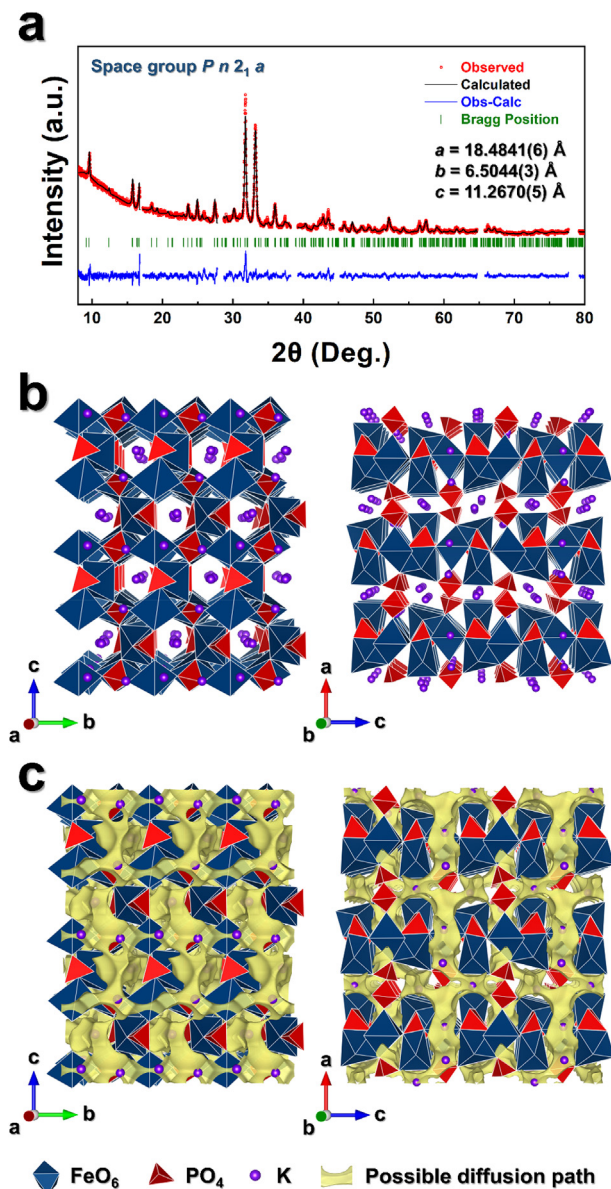


Fig. 2. (a) Rietveld refinement of XRD pattern of $K_4Fe_3(PO_4)_2(P_2O_7)$. (b) Crystal structure of $K_4Fe_3(PO_4)_2(P_2O_7)$. (c) BVS energy map of $K_4Fe_3(PO_4)_2(P_2O_7)$ with all possible K-ion sites in the crystal structure.

$K_4Fe_3(PO_4)_2(P_2O_7)$ structure are (0.4911, 0.758, 0.9872), (0.2816, 0.891, 0.750), (0.3895, 0.432, 0.257) and (0.4746, 0.729, 0.5312).

Using the obtained structural information for $K_4Fe_3(PO_4)_2(P_2O_7)$, we performed first-principles calculations to predict the theoretical properties of $K_4Fe_3(PO_4)_2(P_2O_7)$ in the K-cell system. Various K/vacancy configurations for $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3.5$) were generated using cluster-assisted statistical mechanics (CASM) software. We then calculated and arranged the formation energies of all the $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ configurations (Fig. 3a). The theoretical redox potentials for $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3.5$) were calculated using the following equation:

$$V = \frac{E[K_{x_2}Fe_3(PO_4)_2(P_2O_7)] - E[K_{x_1}Fe_3(PO_4)_2(P_2O_7)] - (x_2 - x_1)E[K]}{(x_2 - x_1)F}, \quad (1)$$

where V is the average redox potential in the $x_1 \leq x \leq x_2$ compositional range, E is the calculated formation energy for the most stable

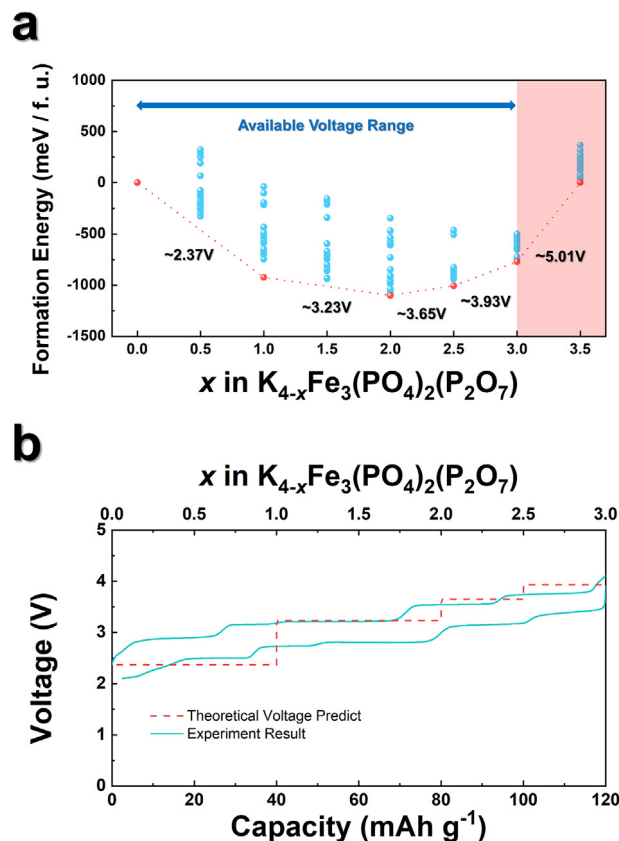


Fig. 3. (a) Formation energy with derived theoretical voltage of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$). (b) Comparison of theoretical voltage prediction and experimental results for $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$).

configuration of each component, and F is the faraday constant. It was predicted that further K^+ deintercalation from $K_1Fe_3(PO_4)_2(P_2O_7)$ corresponding to the Fe^{3+}/Fe^{4+} redox reaction would lead to an excessively high redox potential of ~ 5.01 V (vs. K^+/K). This finding implies that 3 mol K^+ deintercalation per formula unit of $K_4Fe_3(PO_4)_2(P_2O_7)$ arising from the Fe^{2+}/Fe^{3+} redox reaction is available under general cell operation conditions. As shown in Fig. 3b, the predicted redox potential of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$) agrees well with the electrochemically measured charge/discharge curve at C/20 ($1C = 120$ mA g⁻¹) in the voltage range of 2.1–4.1 V (vs. K^+/K), and we confirmed that ~ 3 mol K^+ ions can be reversibly de/intercalated in the $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ structure with a high average operation voltage of ~ 3.21 V (vs. K^+/K).

$K_4Fe_3(PO_4)_2(P_2O_7)$ was tested in a K-cell system using K metal as the counter electrodes and 0.5 M KPF₆ in ethylene carbonate (EC): propylene carbonate (PC): fluoroethylene carbonate (FEC) (20:20:1 by v/v) as the electrolyte. Fig. 4a and b shows the specific capacities of $K_4Fe_3(PO_4)_2(P_2O_7)$ measured at various current rates from C/20 to 5C in the voltage range between 2.1 and 4.1 V (vs. K^+/K). Even at 5C, $K_4Fe_3(PO_4)_2(P_2O_7)$ delivered an exceptionally high specific capacity of ~ 82 mAh g⁻¹, corresponding to $\sim 70\%$ of the theoretical capacity at C/20. We performed first-principles calculation using the nudged elastic band (NEB) method to understand the possible reasons for the outstanding power-capability of $K_4Fe_3(PO_4)_2(P_2O_7)$ (Fig. 4c). Despite the large ionic radius of K^+ , the predicted activation barrier energy for K^+ diffusion is ~ 330 meV along the K_1 – K_2 pathway in the orthorhombic $K_4Fe_3(PO_4)_2(P_2O_7)$ structure, which is similar to that of layered-type cathode materials for LIBs and NIBs [27,28]. This low activation barrier energy of $K_4Fe_3(PO_4)_2(P_2O_7)$ may originate from the large three-dimensional pathways for K^+ diffusion, which enable de/intercalation of K^+ into/out of the host structure at a rate of 5C (Fig. 4a and b). As observed in Fig. 4d, it is notable that $K_4Fe_3(PO_4)_2(P_2O_7)$ exhibited

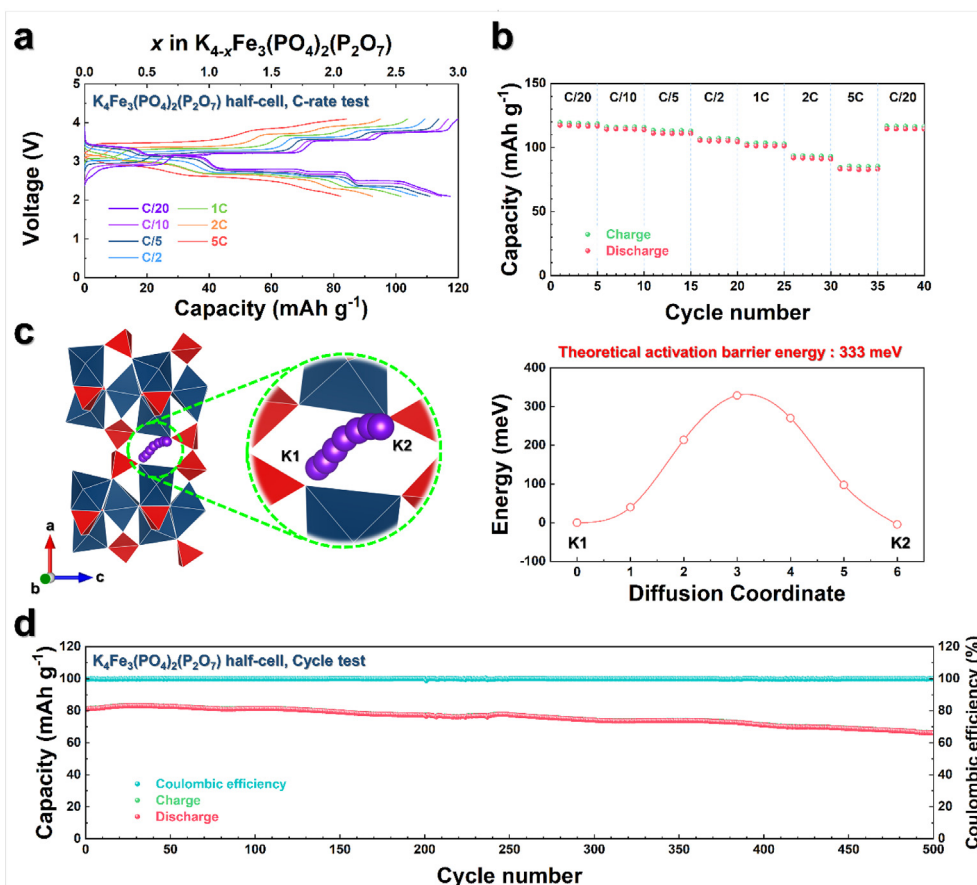


Fig. 4. (a) Charge/discharge curves of $K_4Fe_3(PO_4)_2(P_2O_7)$ in the range of 2.1–4.1 V at various current rates. (b) Power-capability of $K_4Fe_3(PO_4)_2(P_2O_7)$. (c) Diffusion pathway with energy landscape determined using NEB method in $K_4Fe_3(PO_4)_2(P_2O_7)$ structure. (d) Cycling performance of $K_4Fe_3(PO_4)_2(P_2O_7)$ at 5C over 500 cycles.

excellent long-term cycle performances, namely, $\sim 82\%$ capacity retention after 500 cycles at 5C and a Coulombic efficiency of greater than $\sim 99\%$. In addition, it was verified that $K_4Fe_3(PO_4)_2(P_2O_7)$ exhibits not only the capacity retention of $\sim 87\%$ for 200 cycles at C/2 but also the outstanding cycling stability after undergoing different rates (Supporting Fig. S4). The structural stability of $K_4Fe_3(PO_4)_2(P_2O_7)$ was considered to play a pivotal role in the repeated de/intercalation of the large K^+ for long-term cycling. Otherwise, such capacity retention was not possible because of the structural disintegration caused by the dramatic volume change resulting from the repetitive K^+ extraction/insertion processes. Therefore, it is supposed that the aforementioned outstanding cyclability of $K_4Fe_3(PO_4)_2(P_2O_7)$ is associated with the tolerable volume variation during K^+ de/intercalation.

Meanwhile, K metals are more reactive than Li and Na metals, thus usage of K metal anode have a high risk of explosion, caused by external impact or growth of potassium dendrite [29]. Considering the practical use, it is inevitable that introduction of a carbon-based anode electrode material that can accept numerous K^+ , and electrochemical experiment with full-cell system must also be preceded accordingly. Therefore, a full-cell test was conducted with hard carbon anode. Before the full-cell test, hard carbon anode was pre-cycled at the voltage range between 0.001 and 2.0 V, to remove the initial irreversible capacity. $K_4Fe_3(PO_4)_2(P_2O_7)$ /hard carbon full-cell was pre-cycled for 5 cycles at C/3, and tested for 300 cycles at 2C in the voltage range between 2.0 and 4.0 V. Supporting Fig. S5a showed the charge/discharge profiles of $K_4Fe_3(PO_4)_2(P_2O_7)$ /hard carbon full-cell measured at C/3. It delivered a large initial capacity of $\sim 88\ mAh\ g^{-1}$ at C/3. Furthermore, $K_4Fe_3(PO_4)_2(P_2O_7)$ /hard carbon full-cell exhibited remarkable long-term cycling at 2C, retaining over 76% of initial capacity for 300 cycles with high Coulombic efficiency over 99% (Supporting Fig. S5b).

The best of our knowledge, these results showed better performance than the literature reported for the KIB full-cell system [30,31].

To verify the structural change of $K_4Fe_3(PO_4)_2(P_2O_7)$ as a function of K^+ content in the structure, *Operando* XRD analysis of $K_4Fe_3(PO_4)_2(P_2O_7)$ was performed (Fig. 5). The full *Operando* XRD patterns are presented in Supporting Fig. S6. Although overall XRD peaks in the *Operando* XRD patterns are continuously shifted during charge/discharge, it is also observed that the positions and intensities of particular XRD peaks are more highly changed in accordance with the K contents in the structure than those of other XRD peaks, which indicates the complicated phase transitions of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$). As shown in Fig. 5b below, it was verified that the total five phase transitions are occurred at *Operando* XRD patterns of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ during K^+ de/intercalation, which is consistent with the charge/discharge profile of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ under the K-cell system. It was reported that the multiple phase transition of electrode materials for alkali-based ion batteries results in the long-range ordering between alkali-ions and vacancies [32, 33]. Thus, we suppose that the complicated phase transition of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ results from the K^+ /vacancy ordering by the large ionic size of K^+ ions. The lattice parameters calculated by Rietveld refinement of the XRD data indicate that the *c* parameter decreased during K^+ intercalation while the *a* and *b* parameters concurrently increased (Fig. 6a–d), which is consistent with the behavior observed in $Na_4Fe_3(PO_4)_2(P_2O_7)$ in the NIB system [24,25]. It is surprising that the calculated total volume change for $K_4Fe_3(PO_4)_2(P_2O_7)$ was less than 4.4% during K^+ intercalation despite the insertion of K^+ with large ionic size. Therefore, as postulated in the discussion on long-term cyclability (Fig. 4d), the structural integrity supported by the tolerable volume change was verified to be one of reasons for the superior cyclability of $K_4Fe_3(PO_4)_2(P_2O_7)$. In addition, the morphology and XRD pattern of

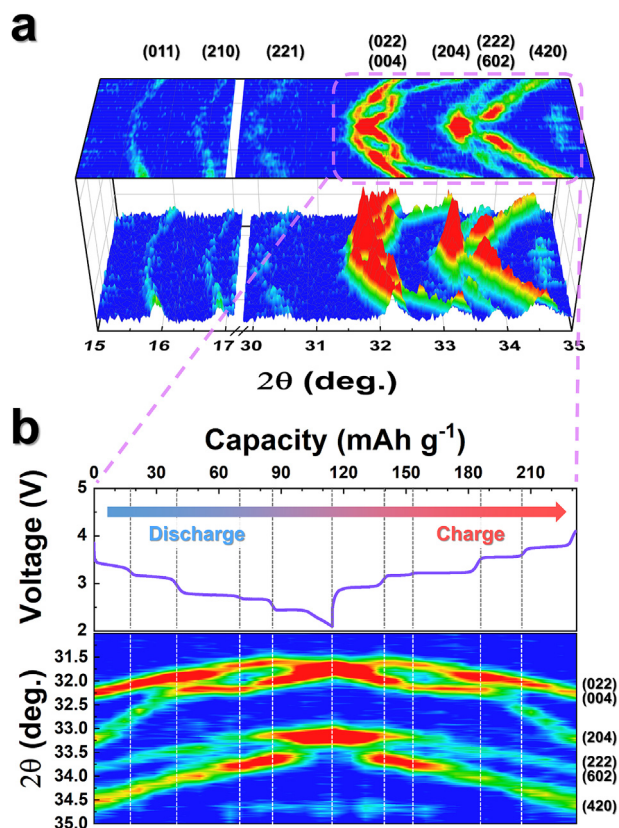


Fig. 5. (a) Operando XRD patterns of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$) (b) Enlarged Operando XRD patterns of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$) with charge/discharge profile.

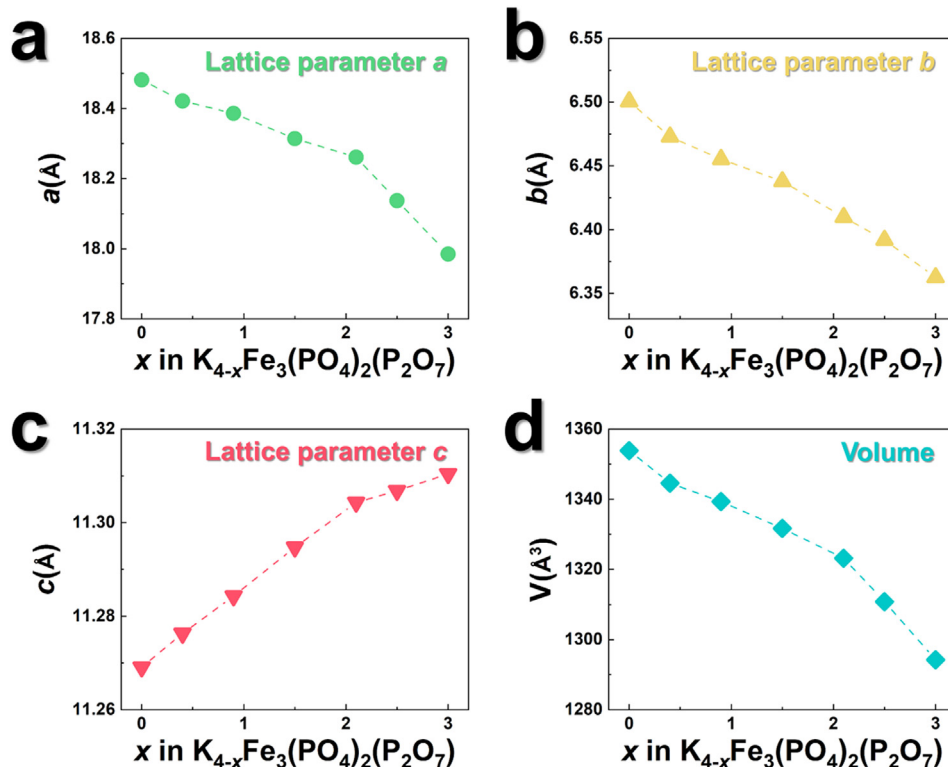


Fig. 6. Change in (a) lattice parameter a , (b) lattice parameter b , (c) lattice parameter c , and (d) volume of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$).

as-prepared $K_4Fe_3(PO_4)_2(P_2O_7)$ electrode were compared with those of the electrode after 500 cycles (Supporting Fig. S7). Attributed to excellent structural stability of $K_4Fe_3(PO_4)_2(P_2O_7)$, severe cracking or structural degradation was not detected at the $K_4Fe_3(PO_4)_2(P_2O_7)$ particle, and its crystal structure was well maintained during the long-term cycling.

The Fe^{2+}/Fe^{3+} redox reaction in $K_4Fe_3(PO_4)_2(P_2O_7)$ during charge/discharge was verified through *ex-situ* XANES analyses. As observed in Fig. 7a, the Fe K-edge was shifted toward a higher energy level during K^+ deintercalation, and then, the Fe K-edge returned to the original energy level after discharging. These results indicate the reversible Fe^{2+}/Fe^{3+} redox reaction for 3 mol K^+ de/intercalation in the $K_4Fe_3(PO_4)_2(P_2O_7)$ structure. In addition, the redox reaction of Fe ions in $K_4Fe_3(PO_4)_2(P_2O_7)$ was confirmed through further analyses using first-principles calculation. Fig. 7b compares the integrated spin moments of Fe ions between $K_4Fe_3(PO_4)_2(P_2O_7)$ and $K_1Fe_3(PO_4)_2(P_2O_7)$. After 3 mol K^+ deintercalation, the total electron spin on Fe ions increased from four to five, which clearly demonstrates the occurrence of the Fe^{2+}/Fe^{3+} redox reaction in $K_4Fe_3(PO_4)_2(P_2O_7)$ to reach ~3.21 V on average in the K-system. In addition, we compared the electrochemical performances among $K_4Fe_3(PO_4)_2(P_2O_7)$ and other cathode materials for KIBs (Supporting Fig. S8), which indicates the outstanding power-capability and energy density of $K_4Fe_3(PO_4)_2(P_2O_7)$ as one of the promising cathode materials for KIBs [34–40]. These findings suggest that $K_4Fe_3(PO_4)_2(P_2O_7)$ is a promising cathode material for KIBs, because of the acceptable high redox potential provided by $Fe^{2+/3+}$ and the sustainability in the structure, enabling the long-term presence of large K^+ .

3. Conclusion

We successfully prepared $K_4Fe_3(PO_4)_2(P_2O_7)$ as a promising cathode for KIBs using an electrochemical ion-exchange process. Using the structural information for $K_4Fe_3(PO_4)_2(P_2O_7)$ verified through XRD with Rietveld refinement, we performed first-principles calculations, and confirmed that 3 mol K ions can be reversibly de/intercalated in the

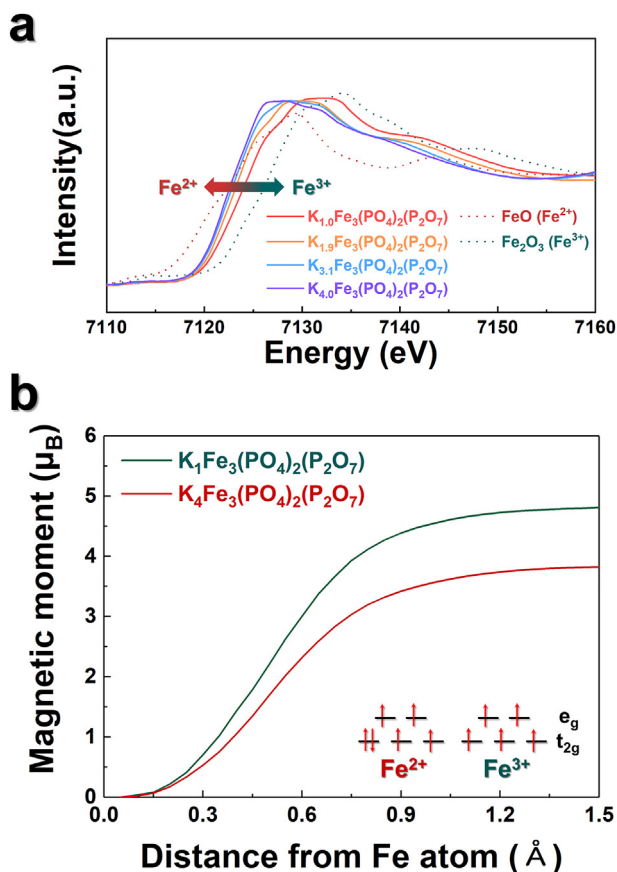


Fig. 7. (a) K-edge XANES spectra of $K_{4-x}Fe_3(PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$). (b) Spin integration of Fe ions in $K_1Fe_3(PO_4)_2(P_2O_7)$ and $K_4Fe_3(PO_4)_2(P_2O_7)$.

$K_4Fe_3(PO_4)_2(P_2O_7)$ structure with a high average operation voltage of ~ 3.21 V (vs. K^+/K). As a result, $K_4Fe_3(PO_4)_2(P_2O_7)$ delivered a large specific discharge capacity of ~ 118 mAh g^{-1} at C/20 ($1C = 120$ mAh g^{-1}) in the voltage range of 2.1–4.1 V (vs. K^+/K), corresponding to its theoretical capacity. Even at 5C, up to $\sim 70\%$ of the theoretical specific capacity was maintained. This superior power-capability of $K_4Fe_3(PO_4)_2(P_2O_7)$ was also demonstrated by the low activation barrier energy of ~ 330 meV despite the migration of K^+ ions with large ionic size. Notably, $K_4Fe_3(PO_4)_2(P_2O_7)$ exhibited exceptionally outstanding cycle-performance with retention of $\sim 82\%$ of the initial capacity after 500 cycles at 5C, which resulted from the low volume change of $\sim 4\%$ during charge/discharge. We believe that our findings will be beneficial for the development of novel low-cost electrode materials with excellent electrochemical properties for KIBs and other rechargeable battery systems.

4. Experimental

4.1. Preparation of $K_4Fe_3(PO_4)_2(P_2O_7)$

$Na_4P_2O_7$ (98%, Duksan Pure Chemicals), $FeC_2O_4 \cdot 2H_2O$ (99%, Sigma Aldrich), $NH_4H_2PO_4$ (98.5%, Sigma Aldrich) and 5 wt% of pyromellitic acid (PA, 96%, Alfa Aesar) were mixed by high-energy ball milling for 12 h and sealed in an Ar-filled glove box. The mixed powder was heated at $300^\circ C$ under Ar flow to remove H_2O . After heating, 5 wt% of PA was added and mixed, and the powder was pelletized. The pelletized sample was reheated to $550^\circ C$ under an Ar/ H_2 mixed-gas (4% H_2) flow. In ion-exchange process, $Na_4Fe_3(PO_4)_2(P_2O_7)$ was assembled to Na-ion cell, and Na ions in the $Na_4Fe_3(PO_4)_2(P_2O_7)$ structure were deintercalated to remove the Na ions, by charging to 4.1 V. After the Na^+ deintercalation, electrode was reassembled to K-ion cell, and K ions were intercalated into

the crystal structure by discharging to 2.1 V. It was subsequently cycled in the voltage range of 2.1–4.1 V for 50 times to remove the residual Na ions.

4.2. Materials characterization

$Na_4Fe_3(PO_4)_2(P_2O_7)$ and $K_4Fe_3(PO_4)_2(P_2O_7)$ were analyzed using XRD (Malvern Panalytical Empyrean) with Cu $K\alpha$ radiation ($\lambda = 1.54178$ Å). The step size was 0.013° , and the 2θ range of 8° – 80° was examined. Rietveld refinement of the XRD data was performed using FullProf software [41]. The morphology and particle size of $Na_4Fe_3(PO_4)_2(P_2O_7)$ and $K_4Fe_3(PO_4)_2(P_2O_7)$ were determined using scanning electron microscopy (SEM; Hitachi SU-8010) and field-emission transmission electron microscopy (FE-TEM; JEOL JEM-F200). The atomic ratios of elements such as Na, K and Fe were determined using inductively coupled plasma optical emission spectrometer (ICP-OES) at Daeduck Analytical Research Institute (DARI). Fe K-edge X-ray absorption spectroscopy (XAS) spectra were obtained at beamline 6D at Pohang Accelerator Laboratory (PAL). The Operando XRD patterns of $K_4Fe_3(PO_4)_2(P_2O_7)$ were analyzed using XRD (PANalytical, Empyrean) with Cu $K\alpha$ radiation ($\lambda = 1.54178$ Å). The step size was 0.026° , and the 2θ range of 15° – 35° was examined.

4.3. Electrochemical characterization

A $K_4Fe_3(PO_4)_2(P_2O_7)$ electrode was prepared by mixing the active material, Super P (conducting carbon) and poly(vinylidene fluoride) (PVDF; binder) using *N*-methyl-2-pyrrolidone (NMP) as the solvent. The total composition of the electrode was adjusted to 80 wt% active materials, 10 wt% Super P, and 10 wt% PVDF. A slurry of the mixture was applied to Al foil using a doctor blade. The electrode was dried at $80^\circ C$ for 12 h to evaporate the NMP. The mass loading of the electrode was ~ 2 mg cm^{-2} .

R2032-type coin cells were prepared using the $K_4Fe_3(PO_4)_2(P_2O_7)$ electrode, K metal as the reference/counter electrode, a separator (Whatman GF/F glass fiber), and 0.5 mol KPF_6 in a EC:PC:FEC in a volume ratio of 20:20:1 as the electrolyte. The coin cells were assembled in an Ar-filled glove box.

Galvanostatic charge/discharge tests were performed at various current rates (C/20, C/10, C/5, C/2, 1C, 2C, and 5C in the range of 2.1–4.1 V) using a battery test system (WonATech WBCS3000). In this experiment, 1C corresponds to ~ 120 mA g^{-1} .

4.4. Fabrication of full-cell

The full cells were fabricated using commercial hard carbon (Kureha) as the anode electrode. The hard carbon was heated at $1000^\circ C$ for 2 h in Ar to remove residual water and air-oxidized substance on the surface of hard carbon particles. The hard carbon electrode was fabricated in the same way and ratio with $K_4Fe_3(PO_4)_2(P_2O_7)$ electrode except Cu foil. To minimize the irreversibility of hard carbon, the hard carbon electrode was pre-cycled as half-cell through direct contact with K metal in the range of 0.001–2.0 V. Finally, R2032-type full-cells were assembled with the $K_4Fe_3(PO_4)_2(P_2O_7)$ cathode and hard carbon anode electrodes (capacity ratio of negative and positive electrodes to be ~ 1.2) in an Ar-filled glove box.

4.5. Computational details

All the density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP) [42]. We used projector-augmented wave (PAW) pseudopotentials [43] with a plane-wave basis set as implemented in VASP. Perdew–Burke–Ernzerhof (PBE) parametrization of the generalized gradient approximation (GGA) [44] was used for the exchange-correlation functional. For DFT calculations, a $2 \times 4 \times 3$ k-point grid was used to calculate a $1 \times 2 \times 1$ supercell structure of $K_4Fe_3(PO_4)_2(P_2O_7)$. The GGA + U method [45] was adopted to

address the localization of the d-orbital in Fe ions, with a U value of 4.3 eV, as used in a previous study on NaFePO_4 [46]. A kinetic energy cutoff of 500 eV were used in all the calculations, and all the structures were optimized until the force in the unit cell converged to within $0.05 \text{ eV } \text{\AA}^{-1}$.

CASM software [47] was used to generate all the K/vacancy configurations for each composition, followed by full DFT calculations on a maximum of 20 configurations with the lowest electrostatic energy for each composition used to obtain the convex hull of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$.

NEB calculations [48] were performed to determine the activation barrier for K diffusion in the $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ structure. To perform the calculations, five intermediate images were generated between each K site. These structures were then calculated using the NEB algorithm with fixed lattice parameters and free internal atomic positions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Hyunyoung Park: Formal analysis, Investigation, Visualization, Writing - original draft. **Hyungsub Kim:** Conceptualization. **Wonseok Ko:** Visualization. **Jae Hyeon Jo:** Visualization. **Yongseok Lee:** Investigation. **Jungmin Kang:** Investigation. **Inchul Park:** Visualization. **Seung-Taek Myung:** Writing - review & editing. **Jongsoo Kim:** Methodology, Writing - review & editing, Supervision.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ensm.2020.02.031>.

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